RAJALAKSHMI ENGINEERING COLLEGE

Tharun Periyan A P 251001289-2025-IT

T2

Started on	Friday, 11 September 2026, 2:24 PM
State	Finished
Completed on	Friday, 11 September 2026, 2:38 PM
Time taken	13 mins 48 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```c
#include <stdio.h>

void qs(int a[], int l, int r) {
    if (l >= r)
        return;

    int i = l, j = r;
    int p = a[(l + r) / 2];

    while (i <= j) {
        while (a[i] < p)
            i++;

        while (a[j] > p)
            j--;

        if (i <= j) {
            int t = a[i];
            a[i] = a[j];
            a[j] = t;

            i++;
            j--;
        }
    }

    qs(a, l, j);
    qs(a, i, r);
}

int main() {
    int n;
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    qs(a, 0, n - 1);

    for (int i = 0; i < n; i++)
        printf("%d ", a[i]);

    return 0;
}
```

```
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```

	Input	Expected	Got	
✔	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✔
✔	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✔
✔	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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